CEPC NOTE

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Sample generation for CEPC

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Abstract

This note focus on the event generation for CEPC studies. The signal and background samples are generated by Monte-Carlo generator Whizard and grouped according to their final states, and the cross asections are given out.

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1 Introduction

The Circular Electron-Positron Collider(CEPC) is a proposed circular electron positron collider experiment to be committed. The main objective is to reconfirm the observation of the Higgs-like particle, which was observed by LHC experiment in July 2012, and study the properties of Higgs boson. The Monte-Carlo event samples play an important role to check the performance of CEPC detector and study the expected branch ratio predicted by SM or BSM of the Higgs boson.

2 Monte Carlo Simulation

The analysis is performed on MC samples simulated on the CEPC conceptual detector, which is based on the International Large Detector (ILD)at the ILC.The event samples at a center-of-mass energy of 240GeV, corresponding to an integrated luminosity of 5.6 ab^{-1} are generated with Whizard 1.95. The simulation is full simulation and slimmed reconstruction samples without hits, and detector geometry is version4.

The cross sections of major Standard Model processes of e^+e^- collisions as functions of center-of-mass energy $\sqrt{s}$, including Higgs production as well as the major backgrounds, are shown in Fig. 1, where the ISR effect has been taken into account. The samples are grouped into signal and background, and then classified according to their final states. The major SM backgrounds, including all the 2-fermion processes($e^+e^- \rightarrow f\overline{f}$, where $f\overline{f}$ refers to all lepton and quark pairs except $t\bar{t}$) and 4-fermion processes(ZZ, WW, ZZorWW, single Z, Single W, Single Z or Single W), which will be discussed in the following sections. Data is stored in $/cefs/data/DstData/CEPC240/CEPC\_v4/$.

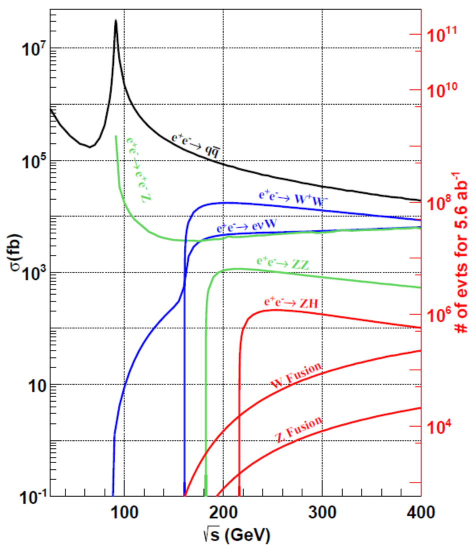

Figure 1: The cross section of major SM processes with ISR effect taken into account

3 Higgs Signal Samples

The Higgs signal samples play an important role in the research of Higgs physics. Production processes for a 125 GeV SM Higgs boson at the CEPC operating at $\sqrt{s}$=240GeV are $e^+e^- \to ZH$ (ZH associate production or Higgsstrahlung), $e^+e^- \to \nu_e\bar{\nu}_e H$ (WW fusion) and $e^+e^- \to e^+e^-H$ (ZZ fusion) as illustrated in Fig 2.

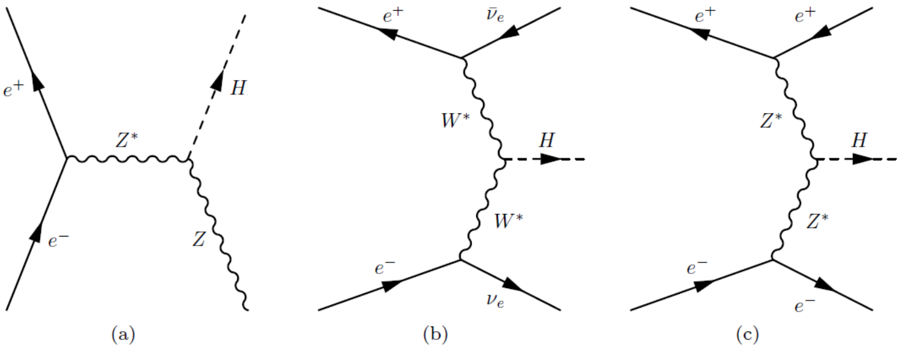

Figure 2: Feynman diagramms of the Higgs boson production at the CEPC:(a)$e^+e^- \to ZH$,(b)$e^+e^- \to \nu_e\bar{\nu}_e H$, and (c)$e^+e^- \to e^+e^-H$.

We mainly focus on the process of $e^+e^- \to ZH$ in which the Higgs and Z bosons will decay further. The Higgs decays inclusively and Z mainly (but not only)decays into visible productions. Because the final states are not specified as Higgs and Z bosons, but their decaying productions instead. In addition, the interference effects between $e^+e^- \to ZH$ and $e^+e^- \to \nu_e\bar{\nu}_e H$ for the $Z \to \nu_e\bar{\nu}_e$ decay and between $e^+e^- \to ZH$ and $e^+e^- \to e^+e^-H$ fot the $Z \to e^+e^-$ decay ,the cross sections of these processes cannot be separated. So the $e^+e^- \to \nu_e\bar{\nu}_e H$ and $e^+e^- \to e^+e^-H$ cross section contributes to the $e^+e^- \to ZH$ process. Table. 1 is the general information about Higgs signal samples.

Table 1: The general information about Higgs signal samples

Process	$Luminosity[ab^{-1}]$	Final states	X-sections(fb)	Events expected	Comments
Higgs signal	5.6	ffH	203.66	1140496	all signals
	5.6	e^+e^-H	7.04	39424	including ZZ fusion
	5.6	$\mu^+\mu^-$H	6.77	37912	
	5.6	$\tau^+\tau^-$H	6.75	37800	
	5.6	$\nu\bar{\nu}$H	46.29	259224	ZH+WW fusion
	5.6	$q\bar{q}$H	136.81	766136	

4 Background Samples

4.1 Two fermions background Samples

Table 2 is the general information about two fermions background samples. In the analysis, we ignore the quark pairs background of $t\bar{t}$. Due to limited storage, data cannot be generated according to standard models and can only be normalized to SM through scale factor.

Table 2: The general information about two fermions background samples

Process	$Luminosity[ab^{-1}]$	Final states	X-sections(fb)	Events generate	Scale factor	Events expected
$e^+e^- \to e^+e^-$	5.6	e^+e^-	24770.90	4000000	346.79%	138717040
$e^+e^- \to \mu^+\mu^-$	5.6	$\mu^+\mu^-$	5332.71	4000000	746.60%	29863176
$e^+e^- \to \tau^+\tau^-$	5.6	$\tau^+\tau^-$	4752.89	4000000	665.40%	26616184
$e^+e^- \to \nu\bar{\nu}$	5.6	$\nu\bar{\nu}$	54099.51	3999999	757.39%	302957256
$e^+e^- \to q\bar{q}$	5.6	$q\bar{q}$	54106.86	9999023	303.03%	302998416

4.2 Four fermions background Samples

In the particles generation automatically, the initial states radiation(ISR) and all possible interference effects are taken into account. Four fermions background samples are divided into ZZ, WW, ZZorWW, single Z, Single W, Single Z or Single W according to its final states. For example, the four fermions in the final states can be combined into two intermediate bosons, respectively two Z bosons or two W bosons, the processes can be named as “ZZ” and “WW” accordingly. Additionally, for the process whose final states contains a pair of electron and the accompanying neutrino, they will be excluded from the ZZ and WW group and named as “single Z” or “single W”, such as including (e, e), (e, ν_e) or (ν_e, ν_e), which indicates from whom the left two fermions come.

Actually some final particles could come from either “ZZ” or “WW”. Those processes will be named as “ZZ or WW ”. The analogous mix also happens between the single Z and single W, which will be called as “single Z or single W ”.

It should be noted that not all end-state particles of the WW process can be combined into two W boson. For example, Fig. 3 is the process ww_h0ccbs. They have the same final particles, but only one can combine into two W boson. If the same final particles also can combine into two Z boson, it's "ZZorWW", such as Fig. 4. The details of the four fermions samples will be discussed in the following sub sections.

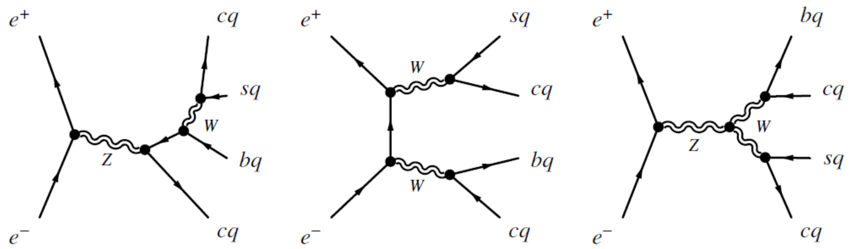

Figure 3: ww_h0ccbs process

4.2.1 ZZ process

Since the Z boson can be considered as a particle composed of fermion and anti-fermion with the same flavor, the particles in the final states of "ZZ" samples are such combinations of particles with same flavor in principle. Furthermore, according to the final state particle types, such as "hadronic", "semi-leptonic", "leptonic", it is divided into three categories, represented by "h","sl","l" respectively. The information of them is shown in Table. 3. The alias used to shorten the table is listed in Table. 4. In Table. 3, nots and notd means not including t,s quarks and t,d quarks. 0 has no physical meaning, it plays a role in segmentation.

Table 3: The general information about zz

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
	zz_h0cc_nots	cq,cq,(dq,bq),(dq,bq)	98.97	499812	110.89%	554232	
zz_h	zz_h0dtdt	down,down,down,down	233.46	1178944	1110.89%	1307376	
	zz_h0utut	up,up,up,up	85.68	432679	110.89%	479808	
	zz_h0uu_notd	uq,uq,(sq,bq),(sq,bq)	98.56	496703	111.11%	551936	
	zz_l04mu	μ^-,μ^+,μ^-,μ^+	15.56	99902	87.22%	87136	
	zz_l04tau	$\tau^-,\tau^+,\tau^-,\tau^+$	4.61	99901	25.84%	25816	
zz_l	zz_l0mumu	$\nu_\tau,\bar{\nu}_\tau,\mu^-,\mu^+$	19.38	99900	108.64%	108528	
	zz_l0taumu	$\tau^-,\tau^+,\mu^-,\mu^+$	18.65	99900	104.54%	104440	6389430
	zz_l0tautau	$\nu_\mu,\bar{\nu}_\mu,\tau^-,\tau^+$	9.61	99900	53.87%	53816	
	zz_sl0mu_down	mu,mu,down,down	136.14	705743	108.80%	762383	
	zz_sl0mu_up	mu,mu,up,up	87.39	448844	109.03%	489383	
zz_sl	zz_sl0nu_down	$nu_{\mu,\tau},nu_{\mu,\tau},down,down$	139.71	708671	110.40	782376	
	zz_sl0nu_up	$nu_{\mu,\tau},nu_{\mu,\tau},up,up$	84.38	429037	110.14%	472528	
	zz_sl0tau_down	tau,tau,down,down	67.31	339928	110.89%	376936	
	zz_sl0tau_up	tau,tau,up,up	41.56	209898	110.88%	232736	

Table 4: The alias for particles

$uq : u,\bar{u}$	$up : u,\bar{u},c,\bar{c}$	$nu_e : \nu_e,\bar{\nu}_e$
$dq : d,\bar{d}$	$down : d,\bar{d},s,\bar{s},b,\bar{b}$	$nu_\mu : \nu_\mu,\bar{\nu}_\mu$
$cq : c,\bar{c}$	$e : e^-,e^+$	$nu_\tau : \nu_\tau,\bar{\nu}_\tau$
$sq : s,\bar{s}$	$mu : \mu^-,\mu^+$	$nu_{\mu,\tau} : \nu_{\mu,\tau},\bar{\nu}_{\mu,\tau}$
$bq : b,\bar{b}$	$tau : \tau^-,\tau^+$	$nu : \nu_{e,\mu,\tau},\bar{\nu}_{e,\mu,\tau}$
$f : e^-,\mu,\tau,\nu_e,\nu_\mu,\nu_\tau,u,d,c,s,b$		q:u,d,c,s,b

4.2.2 WW process

Unlike the Z boson, W boson can be thought to be composed of two quarks with different flavors, therefore the "WW" samples contain the flavor changing processes. And the flavor changing can only occur between the down-type(d,s,b quarks and the antis) and up-type quarks(u,c quarks and the antis). The information of ww is shown in Table 5.

Table 5: The general information about ww

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
ww_h	ww_h0ccbs	cq,cq,bq,sq	5.89	100000	32.84%	32984	50826214
	ww_h0ccds	cq,cq,dq,sq,	170.18	859417	110.89%	953008	
	ww_h0cuxx	cq,uq,down,down	3478.89	17562880	110.93%	19481784	
	ww_h0uubd	uq,uq,bq,dq	0.05	100000	0.28%	280	
	ww_h0uusd	uq,uq,sq,dq	170.45	860029	110.99%	954519	
ww_l	ww_l0ll	$mu, tau, nu_{\mu}, nu_{\tau}$	403.66	2036465	111.00%	2260496	
ww_sl	ww_sl0muq	mu,nu,up,down	2423.43	12238338	110.90%	13571207	
	ww_sl0tauq	tau,nu,up,down	2423.56	12238057	110.90%	13571936	

4.2.3 Single Z process

The final states contains a pair of electron and the accompanying neutrino, they will be excluded from the ZZ and WW group and named as "single Z" or "single W". Single Z are divided into "sze" and "sznu" according to different composition. "sze" contains e^+e^-, and "sznu" contains $\nu_e\overline{\nu}_e$. The information of Single Z is shown in Table 6.

Table 6: The general information about Single Z process

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
single_sze	sze_l0e	uncertain:e^-, e^+, e^-, e^+	78.49	396388	110.89%	439544	9072951
	sze_l0mu	e^-, e^+, μ^-, μ^+	845.81	4270357	11092%	4736536	
	sze_l0nunu	$e^-, e^+, \nu_{\mu,\tau}, \overline{\nu}_{\mu,\tau}$	28.94	146138	110.90%	162064	
	sze_l0tau	e^-, e^+, τ^-, τ^+	147.28	743781	110.89%	824767	
	sze_sl0dd	e,e,down,down	125.83	635351	110.91%	704648	
	sze_sl0uu	e,e,up,up	190.21	960556	110.89%	1065176	
single_sznu	sznu_l0mumu	$\nu_e, \overline{\nu}_e, \mu^-, \mu^+$	43.42	219278	110.89%	243152	
	sznu_l0tautau	$\nu_e, \overline{\nu}_e, \tau^-, \tau^+$	14.57	100000	81.59%	81592	
	sznu_sl0nu_down	$\nu_e, \overline{\nu}_e, down, down$	90.03	454649	110.89%	7504168	
	sznu_sl0nu_up	$\nu_e, \overline{\nu}_e, up, up$	55.59	280749	11088%	311304	

4.2.4 Single W process

Single W process is the process which contain the electron and its accompanying neutrino. The information of Single W is shown in Table 7.

Table 7: The general information about Single W process

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
single_w	sw_l0mu	e,nu_e,mu,$nu_{\mu,\tau}$	436.70	2205350	110.89%	2445520	19517400
	sw_l0tau	e,nu_e,tau,$nu_{\mu,\tau}$	435.93	2201471	110.89%	2441208	
	sw_sl0qq	e,nu_e,up,down	2612.62	13193721	110.89%	14630672	

4.2.5 Single ZorW process

Single ZorW is the process that coincides with both single Z and single W. The information of Single ZorW is shown in Table 8.

Table 8: The general information about Single ZorW process

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
zorw	szeorsw_l0l	$e^-, e^+, \nu_e, \bar{\nu}_e$	249.48	1259867	110.89%	1397088	1397088

4.2.6 ZZorWW process

The final particles can be combined into two Z or two W called ZZorWW, such as Fig 4.

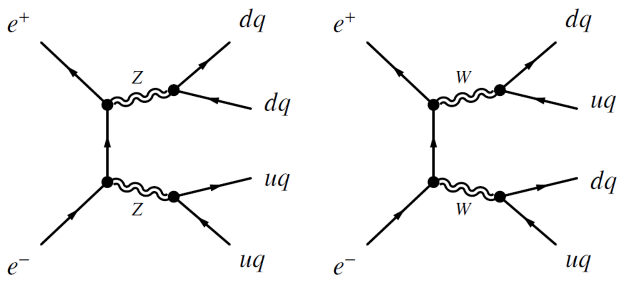

Figure 4: zzorww_h0udud process

The information of ZZorWW is shown in Table 9.

Table 9: The general information about Single ZZorWW process

Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	total
zzorww	zzorww_h0cscs	cq,sq,cq,sq	1607.55	8117636	110.90%	9002280	20440840
	zzorww_h0udud	uq,dq,uq,dq	1610.32	7811146	115.45%	9017792	
	zzorww_l0mumu	$mu, mu, nu_{\mu}, nu_{\mu}$	221.10	1116551	110.89%	1238160	
	zzorww_l0tautau	$tau, tau, nu_{\tau}, nu_{\tau}$	211.18	1066451	110.89%	1182608	

5 Conclusion

At the end of not, all the background information given above is collected in Table 10 and Table 11 . The alias used to shorten the table is listed in Table 4. The principle of abbreviation in Table 11 is detailed in section. 4.

Table 10: The general information about two fermions background samples

Total_name	Process	$Luminosity[ab^{-1}]$	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	Total
2f	$e^+e^- \to e^+e^-$	5.6	e^+e^-	24770.90	4000000	346.79%	138717040	801152072
	$e^+e^- \to \mu^+\mu^-$	5.6	$\mu^+\mu^-$	5332.71	4000000	746.60%	29863176	
	$e^+e^- \to \tau^+\tau^-$	5.6	$\tau^+\tau^-$	4752.89	4000000	665.40%	26616184	
	$e^+e^- \to \nu\bar{\nu}$	5.6	$\nu\bar{\nu}$	54099.51	3999999	757.39%	302957256	
	$e^+e^- \to q\bar{q}$	5.6	$q\bar{q}$	54106.86	9999023	303.03%	302998416	

Table 11: The general information about four fermions background samples

Total_name	Abbreviation	Process	Final states	X-sections(fb)	Events generate	Scale factor	Events expected	Total
single_w	single_w	sw_l0*mu*	e,nu_e,mu,$nu_{\mu,\tau}$	436.70	2205350	110.89%	2445520	19517400
		sw_l0tau	e,nu_e,tau,$nu_{\mu,\tau}$	435.93	2201471	110.89%	2441208	
		sw_sl0qq	e,nu_e,up,down	2612.62	13193721	110.89%	14630672	
single_z	single_sze	sze_l0e	uncertain:e^-,e^+,e^-,e^+	78.49	396388	110.89%	439544	9072951
		sze_l0*mu*	e^-,e^+,μ^-,μ^+	845.81	4270357	11092%	4736536	
		sze_l0nunu	$e^-,e^+,\nu_{\mu,\tau},\bar{\nu}_{\mu,\tau}$	28.94	146138	110.90%	162064	
		sze_l0tau	e^-,e^+,τ^-,τ^+	147.28	743781	110.89%	824767	
		sze_sl0*dd*	e,e,down,down	125.83	635351	110.91%	704648	
		sze_sl0uu	e,e,up,up	190.21	960556	110.89%	1065176	
	single_sznu	sznu_l0mumu	$\nu_e,\bar{\nu}_e,\mu^-,\mu^+$	43.42	219278	110.89%	243152	
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		sznu_sl0nu_up	$\nu_e,\bar{\nu}_e,up,up$	55.59	280749	11088%	311304	
zorw	zorw	szeorsw_l0l	$e^-,e^+,\nu_e,\bar{\nu}_e$	249.48	1259867	110.89%	1397088	1397088
ww	ww_h	ww_h0ccbs	cq,cq,bq,sq	5.89	100000	32.84%	32984	50826214
		ww_h0ccds	cq,cq,dq,sq,	170.18	859417	110.89%	953008	
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	ww_l	ww_l0ll	mu,tau,nu_μ,nu_τ	403.66	2036465	111.00%	2260496	
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zz	zz_h	zz_h0*cc*_nots	cq,cq,(dq,bq),(dq,bq)	98.97	499812	110.89%	554232	6389430
		zz_h0dtdt	down,down,down,down	233.46	1178944	1110.89%	1307376	
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		zz_l0mumu	$\nu_\tau,\bar{\nu}_\tau,\mu^-,\mu^+$	19.38	99900	108.64%	108528	
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	zz_sl	zz_sl0*mu*_down	mu,mu,down,down	136.14	705743	108.80%	762383	
		zz_sl0*mu*_up	mu,mu,up,up	87.39	448844	109.03%	489383	
		zz_sl0nu_down	$nu_{\mu,\tau},nu_{\mu,\tau},down,down$	139.71	708671	110.40	782376	
		zz_sl0nu_up	$nu_{\mu,\tau},nu_{\mu,\tau},up,up$	84.38	429037	110.14%	472528	
		zz_sl0tau_down	tau,tau,down,down	67.31	339928	110.89%	376936	
		zz_sl0tau_up	tau,tau,up,up	41.56	209898	110.88%	232736	
zzorww	zzorww	zzorww_h0cscs	cq,sq,cq,sq	1607.55	8117636	110.90%	9002280	20440840
		zzorww_h0udud	uq,dq,uq,dq	1610.32	7811146	115.45%	9017792	
		zzorww_l0mumu	mu,mu,nu_μ,nu_μ	221.10	1116551	110.89%	1238160	
		zzorww_l0tautau	tau,tau,nu_τ,nu_τ	211.18	1066451	110.89%	1182608	